



NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE FORM 5

SCHOOL YEAR.....

SCHOOL.....

WEEKLY WORKLOAD: 3 periods

TEACHER.....

GRADE.....

TEL.....

Term	Week	Module	Category of action	Competency statement	Lesson title	Objectives	Resources	Nature of lesson		
								Th	Prac	Dig
	1	Network Systems 2	Social networks	Given the need to share experiences and knowledge, learners communicate using platforms that permit interaction in real-time.	Diagnostic evaluation Lesson 1: Notions on social networks	• State the reasons for social networks. • State the characteristics of social networks. • State advantages and disadvantages of online social networks. • Explain the impacts of online social networks.				
					Lesson 2: Using online social networks	• Differentiate between social network and social media. • Differentiate between different online social networks. • Make use of an online social network to share resources and communicate.				
					Lesson 3: Integration activities					
	2	Hardware and Software Systems 3	Exploring Application Areas of Computers	Provided with a situation related to the use of AI, learners write appropriate prompts to get an output or response.	Lesson 4: Introduction to Artificial Intelligence	• Define AI • Explain the history and evolution of AI • Identify and categorise the different types of AI in real world				
					Lesson 5: Applications and Ethics of Artificial Intelligence	• Identify common applications of AI • Explain the uses of AI in different fields • State the ethical issues related to AI				

					Lesson 6: The use of appropriate Prompts to generate AI responses	• Examine prompt results • Discuss prompt results • Write appropriate prompts to solve problems within a given context				
					Lesson 7: Integration activities					
	3				Lesson 8: Notions on organizations and information	• Explain the concepts of information and organization. • Explain the characteristics of good information. • Explain the flow of information within an organization.				
					Lesson 9: Information systems	• Define an information system. • Differentiate manual and automated information systems. • Describe the elements of an information system.				
					Lesson 10: Types of information system	• Describe common types of information system. • Differentiate between batch processing and real-time processing. • Explain how choosing an appropriate process can help ensure the efficiency of an information system.				
	4				Lesson 11: Data capture methods	• Explain the concept of data capture. • Describe common manual and automated data capture methods. • Explain how automated data capture helps improve effectiveness, and reliability of an information system.				
					Lesson 12: Data verification and validation	• Differentiate between data verification and data validation. • Describe data verification and data validation methods. • Explain how data validation and data verification ensure the effectiveness, and reliability of an information system.				
					Lesson 13: Data integrity	• Explain the concept of data integrity. • Explain why ensuring data integrity is important in an information system. • Explain ways of ensuring data integrity in an information system.				
					Lesson 14: Integration activities					
					Evaluation					
	6				Remediation					

		Problem Solving and Coding 3	Developing a system	In a situation of developing a system for a school, business, party management, etc., learners select and design appropriate ways of organising data, describe the processing of data, and select appropriate hardware and software for the given task.	Lesson 15: Stages of SDLC: investigation, analysis, design	- Describe the investigation, analysis and design stages of the SDLC. - Outline the functions of a system to be developed. - Explain the concept of prototyping. - Explain the need for appropriate choice of hardware and software for a system.				
					Lesson 16: Stages of SDLC: development, testing, implementation, maintenance	- Describe the development, testing, implementation, and maintenance stages of the SDLC. - Explain the need for documentation in system development.				
	7				Lesson 17: Implementation strategies	- Describe the different implementation strategies. - Outline the advantages and disadvantages of a given implementation strategy.				
					Lesson 18: Integration activities					
	8	Problem Solving and Coding 3	Designing and implementing simple databases	Provided with a problem in an organisation related to databases, learners build simple database systems for the organisations. The implementation should meet the needs of the organisation.	Lesson 19: Introduction to databases	- Describe common types of database. - Outline advantages and disadvantages of flat file systems. State common features of a DBMS.				
					Lesson20: Relational database design	- Explain the concepts of attribute, record, table, primary key, foreign key, relationship. - Identify attributes, tables, primary key, and relationships in a given situation.				
					Lesson 21: Normalization and Relational models	- Explain the purpose of normalization. - State the properties of a database in 1NF, 2NF, and 3NF. - Identify attributes, tables, primary key, foreign key, and relationships in a given situation.				
	9	Problem Solving and Coding 3			Lesson 22: Use an RDBMS to create tables	- Outline examples of an RDBMS. - Identify common features of a given GUI RDBMS. - Make use of a GUI RDBMS to create tables.				
					Lesson 23: Use an RDBMS to create relationships	- Make use of a GUI RDBMS to create relationships. - Enter data in a table found in a GUI RDBMS. - Perform simple operations such as sorting and filtering data with the aid of a GUI RDBMS.				
					Lesson 24: Use an RDBMS to create queries and reports	- Explain the concepts of database queries and reports. - Make use of a GUI RDBMS to create queries that is coherent to a situation. - Make use of a GUI RDBMS to create reports.				
					Lesson 25: Integration activities					
	10		Managing projects	In a situation where a number of activities (tasks) are given	Lesson 26: Introduction to project management	- Explain the concept of project. - Explain the need for project management.				

				for a project, learners relate the activities in a coherent manner and use project management tools to assemble the tasks in order to finish the project in the shortest time possible.		Describe the phases of the project life cycle.				
					Lesson 27: Project management tools	Describe tools used to plan, monitor and control a project. Produce the Gantt chart for a project.				
					Lesson 28: Project management concepts and metrics 1	- Explain the concept of task, milestone, critical path, slack time, duration of project, critical tasks. - Determine the critical path, slack time, and duration of a project from a Gantt chart				
					Lesson 29: Project management concepts and metrics 2	- Explain the concepts of early start, early finish, late start, and late finish. - Determine the early start, early finish, late start, and late finish in a given situation.				
					Lesson 30: Integration activities					
					Evaluation					
					Remediation					
					Lesson 31: Simple data types	- Explain the concept of data type. - Describe simple data types. - Choose appropriate simple data type for a given situation.				
					Lesson 32: Data structures	- Differentiate between data type and data structure. - Describe the different data structures. Select appropriate data structure for a given situation.				
					Lesson 33: Integration activities					
					Lesson 34: Algorithms to solve common problems 1	- Differentiate between pseudocode and flowchart. - Produce pseudocode and flowchart to solve common problems such as swapping, identification of maximum or minimum.				
					Lesson 35: Algorithms to solve common problems 2	- Differentiate between searching and sorting problems. - Explain the principles of linear search and bubble sort. Produce pseudocode and flowchart to solve common problems such as totalling, counting, ...				
					Lesson 36: Notions on subroutines	- Explain the concepts of subroutines, local variables, global variables, formal parameters, actual parameters. - Identify common parts of a subroutine. - Differentiate between procedures and functions.				
					Lesson 37: Algorithms as subroutines	Produce procedures or functions to solve problems such as totalling, counting, max, min, swap.				
					Lesson 38: Algorithm correctness and efficiency	Explain how dry running can be used to test the correctness of an algorithm.				

						- Describe simple ways of measuring the efficiency of an algorithm. - Establish the correctness and efficiency of a given set of algorithms.				
	15				Lesson 39: Integration activities					
			Converting an algorithm into a program	In a situation where the algorithm for a problem is available, learners develop corresponding instructions in a programming language; using the appropriate IDE.	Lesson 40: Coding 1 (Python)	- State best practices when writing code. - Produce source code for the totalling, counting algorithms.				
					Lesson 41: Coding 2 (Python)	Produce source code for maximum, minimum and swapping algorithms using subroutines.				
					Lesson 42: Coding 3 (Python)	Produce source code for linear search algorithm and bubble sort algorithms.				
	16				Lesson 43: Integration activities					
					Lesson 44: Notions on data encoding	- Explain how data is represented in the computer - Distinguish between data numbers, characters, images and sound - Outline techniques used to encode numbers, characters, images, and sound.				
	17		Representing data in the computer	In a situation where data is stored in the computer, learners identify and use coding schemes to represent data	Lesson 45: Character and positive integers encoding	- Apply a given character encoding scheme to represent a set of characters - Determine the base 2 representation of a positive integer				
					Lesson 46: Addition and subtraction in base 2, 8 and 16	- Identify symbols of base 2, 8 and 16. - Perform addition in any number system - Perform subtraction in any number system				
					Lesson 47: Integration activities					
					Evaluation					
					Remediation					
	18		Describing peripheral devices	In a situation where a computer is to be equipped for various purposes, learners identify appropriate input and output peripheral devices for high productivity.	Lesson 48: Input and output peripherals	- Describe RFID readers, QR code readers, OMR readers, OCR readers, and contactless card readers. - Describe plotters, 3D printer, and actuators. - Choose appropriate input and output peripheral in a given context.				
					Lesson 49: Integration activities					
	19		Creating digital content using a software	Given tasks to accomplish, learners use appropriate features of a software to correctly accomplish the task.	Lesson 50: Using a word processor 1	Produce a formatted word-processed document that makes use of text, tables, graphics, and text boxes				
					Lesson 51: Using a word processor 2	Reproduce a document of at most two pages with the help of a word processor.				
	20				Lesson 52: Solve problems	Reproduce the data and formatting applied to a				

					with spreadsheets 1	spreadsheet • Determine appropriate formulas to meet the different goals in a problem				
					Lesson 53: Solve problems with spreadsheets 2	• Determine appropriate formulas to meet the different goals in a problem • Represent specific spreadsheet data as a given chart • Make use of features of spreadsheets such as sort and filter.				
					Lesson 54: Create presentations	• Create simple presentations of at most 5 slides when given a model of the presentation • Modify a presentation by adding and formatting objects such as images, sound, and videos. • Create animations using a presentation software that meet a given situation.				
					Lesson 55: Create publications	• Produce a given publication when given the model of the publication				
					Lesson 56: Create digital content using generative AI	• Produce word processed documents, presentations, and publications using generative AI tools.				
					Lesson 57: Integration activities					
					Lesson 58: Assistive technology and disabilities	• Explain the concepts of disability, and assistive technology • Describe common assistive technologies for a given disability • Outline features common to operating systems aimed at helping the disabled				
					Lesson 59: Assistive technologies for the elderly	• Explain the importance of assistive technologies for the elderly • Describe technologies that can be used to assist the elderly				
					Lesson 60: Integration activities					
					Evaluation					
					Remediation					
					Lesson 61: Notions on packets	• Explain the concepts packet header, payload, trailer, packet switching. • Describe the structure of a packet. • Describe the USB interface and explain mechanisms it uses to transmit data.				
					Lesson 62: Error detection and packet security	• Explain the need to check for errors after data transmission and how these errors occur. • Explain the need for and purpose of encryption when transmitting data. • Describe error detection methods and how data is				

						encrypted.				
					Lesson 63: Integration activities					
					Lesson 64: Network hardware	• Explain the role of NIC, modem, router, multiplexer, bridge, repeater in a network. Identify hardware needed for internet connectivity in a house, school or organizational network.				
			Setting up LANs	Given a situation that requires setting up a simple network, learners prescribe the appropriate tools and configuration to meet the needs of the situation.	Lesson 65: Network IP configuration	• State the purpose of MAC address and IP address. • Differentiate between IPv4 and IPv6 addresses. • Compare static ip addresses to dynamic ip addresses. • Explain the purpose of configuring ip addresses on a LAN.				
	25				Lesson 66: Integration activities					
					Lesson 67: Notions on security	• Differentiate between computer security, data security, and network security. • Differentiate between vulnerability, threat, and attack. • Identify three security objectives Identify common vulnerabilities linked to data, computers, or networks in a given context.				
			Securing data, computers, and networks.	Given a computing environment where computers, data and networks are exposed to security threats, learners identify potential risks and implement appropriate measures to protect data, computers and networks, justifying the security measures selected.	Lesson 68: Threats and attacks on computer systems	• Identify common threats on data, computers, and networks. • Describe common attacks on data, computers, and networks. • Describe the different types of malware.				
	26				Lesson 69: Data, computer, and network security measures	• Explain measures to secure data. • Explain measures to secure a computer. • Explain measures to secure a network. • Outline common data recovery strategies.				
					Lesson 70: Integration activities					
					Evaluation					
					Remediation					
			Evaluating the impacts of digital identities and digital footprints	Given a situation outlining the online practices of a user, learners evaluate their management of digital footprints, expose the risk they are facing and recommend tips for managing footprints and reputation.	Lesson 71: Notions on digital identities and digital footprints	• Explain the following concepts: digital identity, password, username, identity theft, digital footprint, passive and active digital footprint • Explain characteristics of a good password Explain ways that a user creates passive and active digital footprints				
	27				Lesson 72: Digital identities and footprints management	• Outline best practices to manage multiple digital identities • State positive and negative effects of digital				
	28	Syst em Secu								

						footprints • Explain ways of protecting digital footprints				
					Lesson 73: Integration activities	•				
					Lesson 74: Logic gates	• Identify various logic gates • Describe the different types of logic gates Draw the truth table for a logic gate				
					Lesson 75: Logic circuits and expressions	• Identify logic gates in a circuit or an expression • Produce the truth table or a logic circuit or expression • Produce a logic circuit when given a logic expression				
					Lesson 76: De Morgan's law and Boolean simplification	• State the De Morgan's laws. • Identify situations where De Morgan's law can be used. • State common laws of Boolean simplification.				
					Lesson 77: Integration activities					
					Lesson 78: Storage and processing devices	• Describe processing and storage devices. • Describe the machine instruction cycle.				
					Lesson 79: Other internal components	• State the main ports on the motherboard. • Describe the different types of bus. • State the role of the CMOS and battery.				
					Lesson 80: Integration activities					
					Lesson 81: Notions on the internet	• Explain the concepts of internet, intranet, extranet, protocol. • Outline advantages and disadvantages of the internet. • Explain the role of ISP and browsers in accessing the internet. Describe the importance of URL and its nomenclature.				
					Lesson 82: Notions on digital currency	• Outline common internet services • Explain the concept of digital currencies and how they are used. • State examples of digital currencies. • Explain the process of block chain and how it is used to track digital currency transactions.				
					Lesson 83: Web authoring services	• Outline examples of web authoring tools. • State the role of HTML. • Build a simple web page using HTML and CSS.				
					Lesson 84: Integration activities					

		Hardware and software systems 3	Getting application software	Provided with tasks to produce or edit digital content, learners choose the most appropriate software for each task clearly justifying their choice.	Lesson 85: Types of application software	• Describe common application software for productivity, personal interest, graphics and media, and communication and collaboration. Choose application software for a given situation.					
					Lesson 86: Integration activities						
	33		Describing ways of acquiring and distributing software	Describe ways of acquiring or distributing software	Lesson 87: Methods to obtain software	• Describe the different methods of obtaining software Choose appropriate method to obtain software					
					Lesson 88: Integration activities						
	34	Ethics Society and Legal issues 3	Examine licenses and copyright practices	Placed in a situation with issues related to intellectual property, learners propose ways of identifying free and copyrighted data or digital content, ways of protecting digital content and ways of using content protected by a given license. Proposals should be coherent with the situation and clearly justified.	Lesson 89: Protecting Intellectual property	• Explain the concepts of intellectual property, trademark, patent, copyright, license... • Identify the freedoms provided by a given type of license • State possible national and international consequences of non-respect of intellectual property • Identify exceptions and legal limitations of using and sharing copyrighted content.					
					Lesson 90: Assigning and respecting digital licenses	• Match each type of creative common license to a given permission • Explain ways of identifying free and copyrighted digital content • State platforms where free digital content can be found • State ways of sharing and using digital content legally					
					Lesson 91: Integration activities						
					Evaluation						
	35		Remediation								
	36		Revision								
	END OF PROGRAM										